Green Mobility Management in UAV-Assisted IoT Based on Dueling DQN

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***Abstract*—In most cases, the batteries of sensor nodes in the Internet of Things (IoT) are usually constrained by size and weight, and are difficult to recharge or replace. In traditional wireless sensor networks, data is transmitted in a multi-hop manner, which may cause the high data transmission delay and unbalanced traffic load. In this paper, an Unmanned Aerial Vehicle (UAV)-assisted IoT architecture is introduced, in which UAV is utilized to achieve low-latency and seamless-coverage acquisition of the sensing data. Furthermore, based on the recent advances on deep reinforcement learning algorithms, considering both data delay requirements and network energy consumption, a real-time flight path planning scheme of the UAV in the dynamic IoT sensor networks has been proposed based on dueling deep Q-network (DQN). Besides, the grid-based method is used to handle the network state modeling, which effectively reduces the complexity of the proposed scheme. Simulation results show that the proposed scheme significantly improves the network performance.**

***Index Terms*—UAV-assisted IoT, green mobility management, flight path, deep reinforcement learning, grid-based method**

I. Introduction

As one of the most significant advances in recent years on communication and networking technologies, Internet of Things (IoT) facilitates the interconnection among almost everything for real-time and delay-tolerant information exchange. Specifically, IoT is playing an important role in environment monitoring, agriculture, manufacturing, military scenarios, among others, thanks to its strong capability of collecting the field data for further big data analysis, intelligent learning and decision making in data centers or edge computing devices [1].

The data collection part of IoT suffers from the similar problem as traditional wireless sensor networks: the energy holes caused by the battery drain of IoT nodes which are constrained by size and weight with limited battery capacity. Fortunately, unmanned aerial vehicle (UAV) has attracted much attention and been intensively studied, due to its great convenience on deployment and programming [2][3]. The communication devices loaded on UAVs can be utilized as the base stations or access points with controlled mobility to cover the isolated data islands for effective data uploading, and to increase the life time of the IoT network [4]. By introducing UAVs into IoT and forming UAV-assisted IoT, the network disruption problem can be greatly solved [5].

In the UAV-assisted IoT, UAVs act as mobile sinks (MS), of which patrol path planning becomes one of the key issues affecting the system performance. In [6], the authors propose a stochastic compressive data collection protocol named SMITE for mobile wireless sensor networks, the sensor nodes in the network transmit the data to the aggregation nodes, and the MS moves randomly to collect the data from the aggregation nodes. A grid-based load-balanced routing method (GLRM) has been studied in [7]. The authors adopt the virtual grid-based method, with which the forwarding nodes are elected according to the number of packets carried by the nodes, the Euclidean distance from the center of the grid, and the remaining energy of the nodes. MSs collect the data from the forwarding nodes according to the pre-set path. However, real-time path planning is ignored in both [6] and [7]. To solve this problem, a weighted rendezvous planning (WRP) method is proposed to select the Rendezvous Points (RPs), and the mobile sink only visits the RPs for information collection [8]. The travelling salesman problem (TSP) algorithm is introduced in this work to calculate the shortest path of the MS traversing the RP nodes, thus reducing the data transmission delay. However, issues such as quality-of-service (QoS), computational complexity, energy consumption, etc., are still open and may dramatically affect its application in real scenarios.

On the other hand, deep learning meets reinforcement learning and the resulting major breakthrough in machine learning is already reshaping our world. Deep reinforcement learning (DRL) is widely applied as a generous-purpose algorithm with strong capability of making optimal action decisions and abstracting high-level features from complex scenes [9]. Google DeepMind team has proposed deep Q-network (DQN) in 2015 [10], achieving results beyond human in Atari games. AlphaGo [11] and AlphaGo Zero [12] also achieve a very high level in Go games. After that, DeepMind team made further improvements based on DQN, namely Double DQN [13], which establishes a target DQN that calculates the maximum Q value. Dueling DQN [14] is another major improvement. It divides the Q-value function into two parts, representing the value function of the environment state and the advantage

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function of the action, respectively, avoiding the repeated calculation of the values of the same states, and making the goal of the network learning more clear and stable.

To achieve green UAV-assisted IoT, a layered network architecture is introduced and a real-time UAV path planning scheme is proposed in this paper. The mobility of the UAV is controlled by the moving direction action decisions to minimize power consumption while meeting the delay constraints. The system is formulated as a dynamic decision optimization problem, which is solved by dueling DQN and grid-based method. The main contributions of this paper are as follows.

- The decision sub-layer, the MS sub-layer and the convergence sub-layer are introduced in the UAV-assisted IoT architecture to handle the machine learning, the UAV management and the data aggregation issues, respectively.
- Considering the different data transmission delay constraints from different types of sensing nodes, a priority-based model is adopted. Besides, we use the grid-based method to formulate the system as a discrete one.
- We introduce dueling DQN to optimally and efficiently solve the real-time UAV path planning problem, and use Google TensorFlow as the simulation platform to demonstrate the significant performance improvement.

The rest of this paper is organized as follows. In Section II, the network architecture of the UAV-assisted IoT is described. We present the system model in Section III. Then, in Section IV, the problem of the UAV path planning is formulated as a DRL process. Simulation results are presented and discussed in Section V and conclusions are drawn in Section VI.

II. Network Architecture

A four-layer network architecture is introduced as in Fig. 1 in this section. Sensing layer, access layer, infrastructure layer and application layer devices are connected via wired or wireless channels to provide the internetworking of things, with UAVs as mobile sinks.

A. Sensing Layer

Sensing layer in the network is responsible for sensing and collecting environmental information or field data, composed of massive IoT information acquisition devices such as sensors, cameras, GPS devices, among others.

B. Access Layer

Access layer provides wireless access to the core networks and the data processing services. Different from radio access networks (RANs) in 5G and other cellular networks, the access layer in UAV-assisted IoT systems can be divided into two sub-layers, i.e., convergence sub-layer and MS sub-layer. Collection points (CP) in the convergence sub-layer are selected from the nodes in the sensing layer, responsible for collecting and aggregating the data from the sensing nodes. The aggregated data is then transmitted to the upper MS sub-layer, which is composed of access points with controlled mobility, such as UAVs and unmanned ground vehicles (UGVs). These mobile sinks move around over the CP nodes, based on the paths planned in the real-time manner.

C. Infrastructure Layer

Infrastructure layer in the network architecture is responsible for receiving various data collected by the access layer and uploading the data to the application layer for further processing and storage. Communication satellites, 4G/5G base stations, LoRa base stations, NB-IoT base stations and road-side units (RSUs) are typical devices in the infrastructure layer.

D. Application Layer

User sub-layer in the application layer provides the platform to support various user applications (APPs). The other sub-layer in the application layer, decision sub-layer, facilitates the information processing and data analytics with machine learning and big data techniques. The optimization of the UAV path planning and CP node selecting is also executed in this sub-layer. Note that the algorithms in the decision sub-layer are deployed in the form of APPs as well.

E. Energy Consumption and Delay Issues

The IoT end terminals are generally small-scale devices powered by batteries. Thus, dead devices may result in fault routes and a disruptive network topology. Specifically, keeping the CP nodes in the convergence sub-layer alive is significantly important due to their key function of collecting and aggregating the data from the sensing layer. UAVs in the network also consume much energy, and their power optimization is key to the system performance. Besides, due to the nature of complexity in IoT, different sensors may generate the data packets with different delay constraints. In this case, computing and communication resources could be scheduled with different priorities. Considering both energy consumption

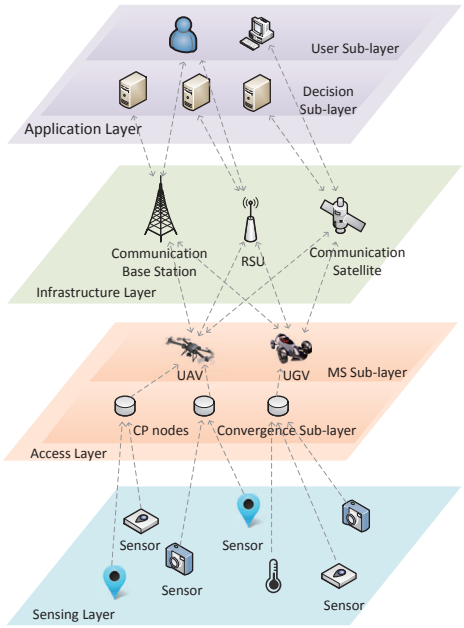

Fig. 1. UAV-assisted IoT architecture.

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and delay issues, in this paper, we aim to achieve the tradeoff by DRL formulation of the system.

III. System Model

The system model including network model and energy model is presented in this section.

A. Problem Description

For simplicity, the single MS scenario is considered in this paper. Real-time path planning for the UAV in the system is studied, taking into account the dynamic delay constraints and the problem of energy holes. The randomly distributed sensor nodes in the sensing layer generate the sensing data packets periodically, and the selected CP nodes upload the collected data to the MS within its communication range. The MS moves based on the path that is planned in the real-time manner, and forwards the data to the application layer via base stations or access points in the infrastructure layer. Path planning decisions are made at the servers in the decision sub-layer and then distributed to the MS. The main objective in this paper is to minimize the power consumption of the CP nodes while meeting the delay constraints, by the means of optimizing of the UAV patrol paths.

B. Network Model

An $N\mathrm{m}\times N\mathrm{m}$ area is considered as the target area with the coverage of the UAV-assisted IoT system. The area is divided into grids. A CP node is selected in each grid to collect the information from the sensors in the grid, and transmit the data to the UAV. Here, direct communication among the nodes in one grid is assumed. Let the center of each grid to be the point that the UAV stays for receiving the data from the CP node. Thus, as shown in Fig. 2, a graph $G(V, E)$ covering the target area can be obtained by connecting all the center points of the grids, where V and E represent the center point of the grid and the edge connecting the points, respectively. Besides, the UAV can change its location to be above the center point of any grid for data collection, along the edge of the graph G with the step length $l(e)$ at each time slot, where $l(e)$ is the length of edge $e \in E$.

C. Energy Consumption Model

The sensors, MS and CPs are wireless devices with limited energy supply. Sensor nodes transmit data to the nearby CP, and thus their energy consumption can be ignored. We only consider the energy efficiency of MS and CPs in this paper.

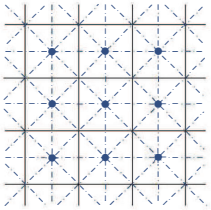

Fig. 2. The area with virtual grids.

In the process of transmission control protocol (TCP) communication, the energy consumption of TCP protocol in the receiving terminal is much smaller than that of the transmitting terminal, so we only consider the energy consumption of data transmission in the receiving terminal. It is assumed that the MS moves with approximately the same distance in each time slot, and this energy consumption is E_{g}. In time slot T_i, define the amount of data collected by the MS from the CPs as k_i, and every continuous N_m time slots as a round. Then, the energy consumption of the MS in a round is

$$E^{\mathrm{MS}} = \sum\nolimits_{i=1}^{N_{\mathrm{m}}} (E_{\mathrm{e}} \times k_i + E_{\mathrm{g}}), \tag{1}$$

where the energy consumption of the MS for collecting each data unit from the CPs is E_{e}. According to the definition of energy efficiency in [15], the energy efficiency of MS can be written as

$$\gamma(k_i) = \frac{\sum_{i=1}^{N_{\mathrm{m}}} k_i}{E^{\mathrm{MS}}}. \tag{2}$$

Consequently, more data $K = \sum_{i=1}^{N_{\mathrm{m}}} k_i$ collected in every N_{m} time slots by the MS contributes to the higher the energy efficiency.

The communication energy consumption of the CP is composed of the energy for collecting data from sensor nodes, represented by $E_{\mathrm{R}}^{\mathrm{CP}}$, and the energy for transmitting the data to the MS, represented by $E_{\mathrm{T}}^{\mathrm{CP}}$. We have

$$E^{\mathrm{CP}} = E_{\mathrm{R}}^{\mathrm{CP}} + E_{\mathrm{T}}^{\mathrm{CP}}. \tag{3}$$

Similarly, at time instance t, the energy consumption of the CP receiving k_j bits from the sensor nodes in the same grid is

$$E_{\mathrm{R}}^{\mathrm{CP}} = E_{\mathrm{r}}^{\mathrm{CP}} \times k_j. \tag{4}$$

TCP protocol is adopted in the data transmission between the CPs and the MS. TCP is connection-oriented, and its communication process can be basically divided into three parts: data processing and encapsulation, establishing connection, and releasing connection. Then, in each communication process with TCP, the fixed energy consumption of the transmitting terminal is

$$E^{\mathrm{TCP}} = E_{\mathrm{h}} + E_{\mathrm{c}} + E_{\mathrm{r}}, \tag{5}$$

where E_{h}, E_{c}, and E_{r} are energy consumptions for data processing and encapsulation, establishing connections, and releasing connections, respectively.

At time instance t, if the CP sends data to the MS, the energy consumption of the CP is

$$E_{\mathrm{T}}^{\mathrm{CP}} = E_{\mathrm{t}}^{\mathrm{CP}} \times k_j + E^{\mathrm{TCP}}, \tag{6}$$

where $E_{\mathrm{t}}^{\mathrm{CP}}$ is the unit energy consumption of the CP in the process of the data transmission based on TCP. Due to the unstable channel state in the IoT environment, we have

$$E_{\mathrm{t}}^{\mathrm{CP}} = S_{\mathrm{T}}/R_{\mathrm{CP-MS}}, \tag{7}$$

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where

$$S_{\mathrm{T}} = S_{\mathrm{R}}/h_j, \tag{8}$$

and

$$R_{\mathrm{CP-MS}} = W\log_2(1 + S_{\mathrm{R}}/n_0). \tag{9}$$

Here, S_{T} represents the transmission power of the CP, S_{R} is the receiving power of the MS, h_j is the channel attenuation factor, $R_{\mathrm{CP-MS}}$ represents the fixed data rate between the CP and the MS, W represents the channel bandwidth, and S_{R}/n_0 is the SNR.

Then, at time instance t, the energy consumption of the communication between the CP and the MS is

$$E^{\mathrm{CP}} = E_{\mathrm{r}}^{\mathrm{CP}} \times k_j + \frac{S_{\mathrm{R}}/h_j}{W\log_2(1 + S_{\mathrm{R}}/n_0)} \times k_j + (E_{\mathrm{h}} + E_{\mathrm{c}} + E_{\mathrm{r}}). \tag{10}$$

The energy efficiency model of the CP is

$$\varepsilon(k_j, h_j) = \frac{k_j}{E^{\mathrm{CP}}}. \tag{11}$$

It can be observed from (11) that the energy consumption of the CP is related to the channel attenuation factor h_j and the amount of transmitted data k_j, and the larger h_j and k_j improves the energy efficiency.

IV. DRL-Based MS Path Planning

In this section, the system is formulated as a DRL problem. The MS is considered as an agent. The following assumptions are taken.

- We only consider the data collection of the MS at the center of the grid, ignoring the data collected when moving.
- In the same grid, the most stringent delay of the data is used as the delay of the grid.

A. Data Prioritization

In order to ensure the immediate transmission of data with the most strict delay constraint in the case of limited transmission resources at the MS, the data is prioritized according to the data delay constraints and the energy consumption. Data should be collected and uploaded within the constraints. Consequently, the data packets whose delay is close to its delay constraint should have higher delay priority. Thus, the delay priority is written as

$$\alpha(t) = \frac{k_j}{T_{\mathrm{m}} - t}, \tag{12}$$

where T_{m} is the latency constraint for the data.

According to (2) and (11), the energy consumption depends on the amount of data k_j collected by the MS in each time slot, the amount of data k_j sent by the CP node to the MS, and the channel state h_j. The energy consumption priority is defined as

$$\beta(k_i, k_j, h_j) = A_1 \cdot \gamma(k_i) + A_2 \cdot \varepsilon(k_j, h_j), \tag{13}$$

where A_1 and A_2 are weight coefficients. Then, we define the priority function as

$$P(t, k_i, k_j, h_j) = B_1 \cdot \alpha(t) + B_2 \cdot \beta(k_i, k_j, h_j), \tag{14}$$

where B_1 and B_2 are weight coefficients. Furthermore, the priority function values are equally divided into L segments to obtain the grids of different priorities $p = \{p_0, p_1, p_2, \cdots, p_l\}$, where the priority function values $P_l \in \left(P_l^{\min}, P_l^{\max}\right]$. Here $P_l^{\min}$ and $P_l^{\max}$ represent the minimum and maximum priority function values of the priority p_l, respectively.

B. System State

To utilize the DRL model, we map the grid network model to an RGB image, with different colors representing different priorities. In each time interval T during the MSs one-step movement, some data is generated at a random location, and the total amount of data in the network remains the same roughly. Therefore, the state of the system changes for every time interval T, and the system state is

$$s(t) = s(nT) = \begin{bmatrix} s_{1,1}^n & \cdots & s_{1,N}^n \\ \vdots & \ddots & \vdots \\ s_{N,1}^n & \cdots & s_{N,N}^n \end{bmatrix} = (e_0, e_1, \cdots, e_n), \tag{15}$$

where $s_{p,q}^n = c_k (k = 0, 1, \cdots, m)$ represents the priority in each grid. Consequently, the network has $e_0, e_1, \cdots, e_n$, $n+1$ possible states.

C. Action Space

In the UAV-assisted IoT, action $a(t)$ is the direction that the MS moves at t. As shown in Fig. 3, the action space of the system is defined as A,

$$a(t) = \{m | m \in A\}. \tag{16}$$

D. Reward Function

The reward function is the evaluation when taking an action with a certain state in reinforcement learning. In this paper, the priority of the data packet in the grid received by the MS after taking the action is adopted as this evaluation.

At time instance t, the reward function is defined as a piecewise function according to the priority $p_{a_t} = \{p_{a_t,0}, p_{a_t,1}, p_{a_t,2}, \cdots\cdots, p_{a_t,l}\}$ of the data in the grid where the agent is located after the action $a(t)$ is taken. It can be represented as

$$R(a_t) = r_l, p_{a_t,l}, \tag{17}$$

where $r_l = \frac{P_l^{\min} + P_l^{\max}}{2}$. Higher data priority of the grid received by the MS contributes to higher the reward value. Then, the optimization goal of the system is written as

$$\begin{aligned} &\max \sum\nolimits_{t \in [0, nT]} R(a_t), \\ &\text{s.t.} \quad a_t \in A. \end{aligned} \tag{18}$$

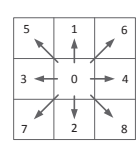

Fig. 3. System action space.

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Algorithm 1: Dueling DQN algorithm in UAV-assisted IoT.

Initialization.
Initialize parameters of main DQN: $\omega = \theta$.
Initialize parameters of target DQN: $\omega' = \theta$.
Initialize the experience buffer M.
Initialize the probability of random actions e.
Training.
for $j \in [J]$ **do**
  Reset environment state s.
  for $t \in [T]$ **do**
    Randomly select a probability $p \in (0, 1)$.
    if $p \leq e$ **then**
      Randomly select an action: $a_t \in A$.
    else
      $a_t = \arg\max_a Q(s, a; \omega)$.
    end if
    Execute the action a_t.
    Obtain the experience (s_t, a_t, r_t, s_{t+1}) and store them into the experience buffer M.
    Get τ samples from the experience buffer M.
    Calculate the target Q value using
    $Y_{\text{target}} = R_{t+1}$
    $+\gamma Q_{\text{target}}(s_{t+1}, \arg\max_a(Q(s_{t+1}, a_{t+1}; \omega), \omega'))$.
    Calculate the loss value by the loss function
    $L(\omega) =$
    $\mathrm{E}[(r_t + \gamma \max_a Q(s_{t+1}, a_{t+1}; \omega') - Q(s_t, a_t; \omega))^2]$.
    Update parameters with Adam Optimizer.
    Update e.
  end for
end for

E. Path Planning Algorithm Based on Dueling DQN

With the formulation of system states, action space and reward function of the system, we use dueling DQN to train the MS for moving path planning. In the system formulated in this paper, the change of the MSs location and the transmission/generating of the sensing data causes the transition of the system state in each time slot. To capture this behavior of the system, the experience replay mechanism [16] is adopted. Massive exploration is performed by the MS, and then in each time slot, the system state s, the action a, the obtained reward value r, and the next state s' after taking the action a are stored in the memory as (s, a, r, s'). A sample set D is also formed, and samples are extracted randomly to update the network. The process is as shown in Algorithm 1.

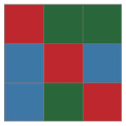

Fig. 4. RGB image based on network states.

TABLE I
SIMULATION PARAMETERS

Parameters	Values
Main DQN learning rate	0.0001
Target DQN learning rate	0.001
Discount factor	0.99
Update frequency	4
Pre-training steps	20000
Total steps	50000
Number of steps per round	100

V. SIMULATION RESULTS AND DISCUSSIONS

In this section, the simulation results of the mobility management scheme in green UAV-assisted IoT is presented and analyzed, comparing with the existing classical mechanisms including random path and fixed path planning. The Nm$\times N$m area is divided into 100 grids. According to the data prioritization presented before, the grid data is divided into 6 levels, $P_{a_t} = \{P_{a_t,0}, P_{a_t,1}, \cdots\cdots, P_{a_t,5}\}$. The reward values are also set according to (17). We use Ubuntu 16.04 as the operating system, and TensorFlow 1.0.0 as platform for our simulation to implement dueling DQN.

Based on the grid and the data priority, the system state is mapped to an RGB image as shown in Fig.4, different colors representing different priorities. Simulation parameters are shown in Table I.

Fig.5 shows the performance of the MS with different random moving probabilities. The MS generates samples for learning during its movement, with the decreasing probability of random movement and the increasing probability of moving according to the generated action from the learning network. As the number of iterations grows, the probability of random movement reduces gradually to a predetermined value e. As we can see from this figure, lower e provides better system performance with the cost of slower convergence. If we use $e = 0$, the network converges after about 25500 iterations with the converged performance evaluation 60.5. If $e = 0.1$ is adopted, 15000 iterations brings the converged performance

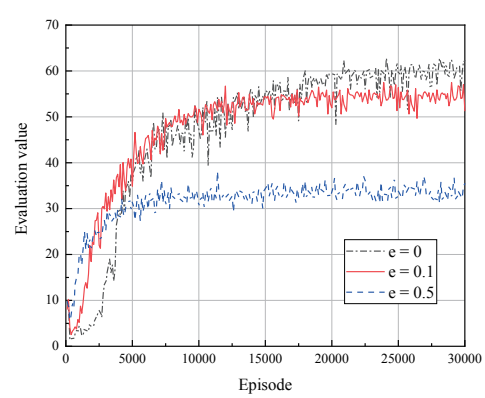

Fig. 5. Performance evaluation with different random motion probabilities.

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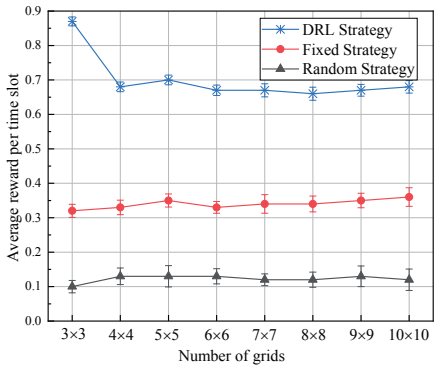

Fig. 6. Comparison of system performance with different grid numbers.

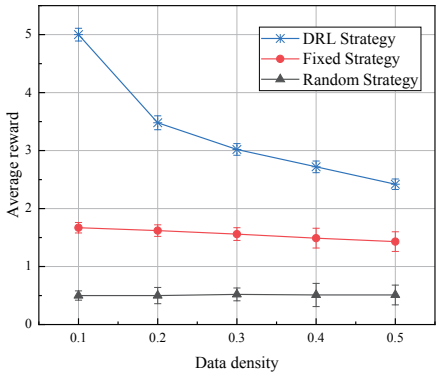

Fig. 7. Average system reward with different network density.

value 54.8.

In Fig. 6, we compare the average reward in all time slots of the three schemes with different number of grids but the same data density. Here, data density is defined as the probability of grids with data to transmit in the network. It is obviously observed that the average reward of DRL-based real-time path planning scheme is obviously higher than the others with the same number of grids, indicating that the DRL-based intelligent agent learns the data priority in different grids. Besides, this figure also shows that the number of grids is not the key factor affecting the system performance dramatically. We also draw the confidential interval in this figure with the confidential level 90%. It can observed that most results are close to the average value.

Furthermore, the system performance of our proposed scheme depends on data density, which can be seen in Fig. 7. As the increasing of data density, the DRL-based scheme shows significant performance degradation. This is because higher data density usually results in larger state space, which brings too much computational complexity and may seriously affect the learning performance. The other two existing schemes do not suffer from high data density. However, our scheme always performs much better than the existing ones with the confidential level 90%, even when the data density is high.

VI. Conclusions

In this paper, considering the random distribution of sensing nodes and the resulting unbalanced traffic load in UAV-assisted IoT systems, a green and real-time MS path planning scheme has been proposed, with the objective of reducing power consumption while meeting the delay constraints for different data types. The grid-based method was adopted to formulate UAV path planning as a dueling DRL problem with reduced state space and computational complexity. Extensive simulation results shown in this paper indicate the significant performance improvement of the proposed scheme compared to existing ones.

References

[1] J. Yick, B. Mukherjee, and D. Ghosal, "Wireless sensor network survey," *Computer Networks*, vol. 52, no.12, pp. 2292-2330, Aug. 2008.

[2] P. Si, F. R. Yu, R. Yang, and Y. Zhang, "Navigation Data-Assisted Opportunistic Spectrum Scheduling for Network-Based UAV Systems: A Parallel Restless Bandits Formulation," *Wireless Personal Communications*, vol. 85, no. 1, 2015.

[3] N. H. Motlagh, T. Taleb, and O. Arouk, "Low-Altitude Unmanned Aerial Vehicles-Based Internet of Things Services: Comprehensive Survey and Future Perspectives," *IEEE Internet of Things Journal*, vol. 3, no. 6, 2016.

[4] Y. Zeng, and R. Zhang, "Energy-Efficient UAV Communication With Trajectory Optimization," *IEEE Transactions on Wireless Communications*, vol. 16, no. 6, pp. 3747-3760, 2017.

[5] V. Sharma, M. Bennis, and R. Kumar, "UAV-Assisted Heterogeneous Networks for Capacity Enhancement," *IEEE Communications Letters*, vol. 20, no. 6, pp. 1207-1210, 2016.

[6] L. Guo, R. Beyah, and Y. Li, "SMITE: A stochastic compressive data collection protocol for Mobile Wireless Sensor Networks," in *IEEE INFOCOM*, 2011.

[7] Q. Liu, K. Zhang, J. Shen, Z. Fu, and N, Linge, "GLRM: An improved grid-based load-balanced routing method for WSN with single controlled mobile sink," in *Proceedings of the 18th International Conference on Advanced Communication Technology(ICACT)*, 2016.

[8] H. Salarian, K. W. Chin, and F. Naghdy, "An Energy-Efficient Mobile-Sink Path Selection Strategy for Wireless Sensor Networks," *IEEE Transactions on Vehicular Technology*, vol. 63, no. 5, pp. 2407-2419, 2014.

[9] Q. Liu, J. Zhai, Z. Zhang, et al, "A survey on Deep Reinforcement Learning," *Chinese Journal of Computers*, 2018.

[10] V. Mnih, K. Kavukcuoglu, D. Silver, et al, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, 2015.

[11] D. Silver, A. Huang, C. J. Maddison, et al, "Mastering the game of Go with deep neural networks and tree search," *Nature*, vol. 529, no. 7587, pp. 484-489, 2016.

[12] D. Silver, J. Schrittwieser, K. Simonyan, et al, "Mastering the game of Go without human knowledge," *Nature*, vol. 550, no. 7676, pp. 354-359, 2017.

[13] H. V. Hasselt, A. Guez, and D. Silver, "Deep Reinforcement Learning with Double Q-learning," [Online]. Available: http://arxiv.org/abs/1509.06461v1

[14] Z. Wang, T. Schaul, M. Hessel, et al, "Dueling network architectures for deep reinforcement learning," *arXiv preprint arXiv:1511.06581*, 2015.

[15] W. Jiang, and G. Cheng, "Optimization and quantitative evaluation mechanism of energy efficiency model on wireless sensor networks," *Journal of Harbin Institute of Technology*, vol. 46, no. 5, pp. 87-94, 2014.

[16] L. J. Lin, "Reinforcement learning for robots using neural networks," Ph.d.thesis Carnegie Mellon University, 1993.

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